

MLAWA, Poland

WMO: 122700

Lat: 53.1042N

Lon: 20.3611E

Elev: 149

StdP: 99.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-13.4	-18.6	0.7	-15.8	-15.8	1.0	-12.8	12.0	4.3	10.6	3.5	2.5	80	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	29.5	20.1	27.6	18.9	25.8	18.2	21.2	27.4	20.2	25.5	19.2	24.0	3.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.2	23.3	18.3	13.4	22.3	17.3	12.6	21.5	61.9	27.6	58.5	25.6	55.2	24.1	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.4	8.2	7.2	DB	-18.6	32.7	3.8	1.7	-21.3	33.9	-23.5	34.9	-25.7	35.9	-28.4	37.1	(4)
(5)			WB	-18.9	22.8	3.7	1.0	-21.5	23.5	-23.7	24.1	-25.7	24.7	-28.4	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-2.8	-1.4	2.3	8.4	13.2	16.3	18.7	18.3	13.4	8.1	3.2	-0.9
	DBStd	8.66	5.47	5.06	4.36	4.41	4.09	3.57	3.42	3.22	3.41	3.87	4.00	5.10
	HDD10.0	1684	397	319	241	80	15	1	0	0	8	83	204	337
	HDD18.3	3844	655	553	498	298	166	82	38	41	151	316	453	595
	CDD10.0	1002	0	0	1	33	115	189	270	257	111	26	1	0
	CDD18.3	121	0	0	0	1	7	20	49	39	4	0	0	0
	CDH23.3	1077	0	0	0	12	74	190	424	337	39	0	0	0
	CDH26.7	266	0	0	0	1	8	41	122	86	8	0	0	0
Wind	WSAvg	3.4	3.9	3.8	3.9	3.5	3.2	3.1	3.0	2.8	3.0	3.3	3.6	3.8
Precipitation	PrecAvg	568	36	30	34	36	56	66	76	58	53	41	42	40
	PrecMax	701	106	75	73	105	152	172	191	123	195	114	114	82
	PrecMin	411	3	7	10	3	27	16	16	14	8	6	8	9
	PrecStd	72	20	15	15	21	25	33	37	27	43	29	20	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	9.6	16.0	25.1	27.8	30.6	32.4	31.9	27.6	20.4	13.2	10.0
		MCWB	6.9	7.4	9.8	15.4	18.2	21.0	21.0	20.7	18.2	15.2	11.4	9.1
	2%	DB	6.5	8.0	13.1	21.4	25.5	27.5	29.8	28.9	23.9	17.8	11.2	8.1
		MCWB	5.6	6.5	8.5	13.3	17.3	19.0	20.3	19.8	16.6	14.0	9.9	7.2
	5%	DB	5.0	6.5	10.7	18.5	23.3	25.4	27.7	26.9	21.6	15.7	9.7	6.7
		MCWB	4.3	5.2	7.4	11.7	16.0	17.8	19.2	18.5	15.7	12.9	8.7	5.9
	10%	DB	3.6	4.8	8.5	16.1	20.9	23.5	25.6	24.9	19.6	13.9	8.5	5.3
		MCWB	2.9	3.7	6.1	10.4	14.7	16.9	18.4	17.9	14.8	11.9	7.6	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	8.0	11.1	16.3	19.6	22.0	22.5	22.2	19.3	16.2	11.8	9.2
		MCDB	7.9	8.9	14.8	24.5	25.9	28.6	29.0	29.0	25.4	19.2	12.7	9.7
	2%	WB	5.7	6.6	9.2	14.1	18.1	20.2	21.2	20.9	17.7	14.5	10.1	7.3
		MCDB	6.4	7.8	12.0	19.9	23.9	25.7	27.5	26.8	22.1	16.8	11.1	8.0
	5%	WB	4.3	5.3	7.7	12.4	16.8	18.9	20.3	19.8	16.5	13.3	8.8	6.0
		MCDB	5.0	6.4	10.2	17.3	22.0	23.8	25.8	24.8	20.2	15.3	9.5	6.7
	10%	WB	3.0	3.8	6.2	10.9	15.5	17.7	19.3	18.8	15.4	12.1	7.7	4.7
		MCDB	3.5	4.7	8.3	14.9	19.8	22.0	24.0	23.2	18.7	13.8	8.4	5.3
Mean Daily Temperature Range	5% DB	MDBR	4.3	5.1	7.0	9.5	10.3	10.3	10.1	10.4	9.0	7.1	4.3	4.1
		MCDBR	4.4	6.2	10.5	13.3	13.0	13.0	12.9	13.1	12.2	9.6	5.9	4.6
		MCWBR	4.0	5.1	6.8	6.9	6.4	6.1	5.1	5.4	5.9	6.1	4.9	4.2
	5% WB	MCDBR	4.4	6.1	9.7	12.1	11.9	11.6	11.2	11.3	10.7	8.8	5.6	4.5
		MCWBR	4.1	5.1	6.8	6.6	6.4	6.1	5.2	5.4	5.7	5.9	4.9	4.2
Clear-Sky Solar Irradiance	taub	0.306	0.337	0.369	0.417	0.406	0.402	0.435	0.425	0.383	0.360	0.346	0.304	
	taud	2.226	2.234	2.238	2.162	2.253	2.295	2.222	2.247	2.340	2.386	2.298	2.263	
	Ebn at Noon	654	751	806	810	843	850	811	795	787	724	602	584	
	Edh at Noon	65	88	108	133	128	124	131	121	98	76	62	53	
All-Sky Solar Radiation	RadAvg	0.65	1.35	2.58	4.12	5.24	5.49	5.09	4.42	3.14	1.69	0.66	0.44	
	RadStd	0.12	0.22	0.39	0.54	0.52	0.49	0.58	0.40	0.44	0.28	0.07	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-163	+60	N/A
Regional (28 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	-155	+74	N/A

Nomenclature: See separate page